

3.4 Differentiability

A concept of fundamental importance for the upcoming sections is the notion of *differentiability*. In Calculus I this concept is discussed in detail in world of 2-space. However, we will need to extend the idea of what we mean by differentiability in Calculus I to better fit the broader context of 3-space.

Motivation:

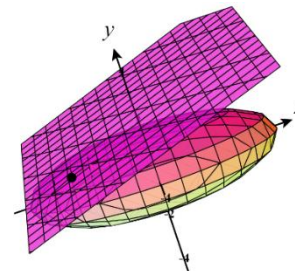
Recall (From Calculus I): A function $f(x)$ is differentiable at x_0 if $f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{h}$ exists.

Differentiability in Calculus I was an important concept because it implies several other important properties about $f(x)$:

- (1) $f(x)$ has a non-vertical tangent line at x_0 .
- (2) $f(x)$ may be approximated by a linear function near x_0 .
- (3) $f(x)$ is continuous at x_0 .

In this section we want to *extend* this notion differentiability to functions of two or three variables such that:

- (1) $z = f(x, y)$ has a nonvertical tangent plane at the point $(x_0, y_0, f(x_0, y_0))$.
- (2) $z = f(x, y)$ can be approximated by a linear function for points near (x_0, y_0) .
- (3) $z = f(x, y)$ is continuous at (x_0, y_0) .



The following definition allows for our desired extension to take place. Details of how this concept may be explored/discovered are shown in more detail in the notes on Blackboard.

Definition: A function f of two variables is said to be differentiable at (x_0, y_0) provided $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ both exist and

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{\Delta f - f_x(x_0, y_0)\Delta x - f_y(x_0, y_0)\Delta y}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = 0$$

Note: A similar extension exists for functions of three variables. To see an explanation of the derivation of this idea consult page 942 of the textbook.

Note: Intuitively, this means that the linearization of f at (a, b) does a good job of approximating f near (a, b) . As such, we can understand that a surface being differentiable loosely mean that the tangent plane exists and does a nice job of modeling the surface near the point of tangency.

Note: The expressions subtracted in the numerator may be seen as a multiplication of matrices', which will become of great importance if one wishes to further extend the notion of differentiability to n -dimensional spaces. We will see how this idea of matrices can be used when we discuss the Jacobian in section 4.7.

Explanation of above Definition (Not done in class)

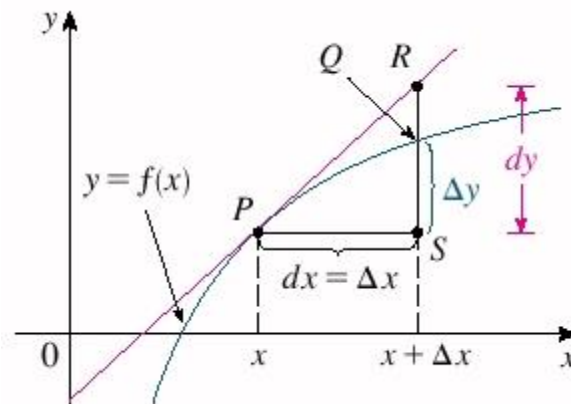
Recall (Differentials from Calculus I):

The geometric meaning of a differential can be seen in the following diagram.

Suppose $y = f(x)$ and P is a point on the graph of this function with coordinates: $(x, f(x))$.

Let Q be the point on the graph of this function with coordinates $(x + \Delta x, f(x + \Delta x))$.

Then, we think of dx as Δx (the change in x). Mathematically, $dx = \Delta x$.



In Calculus I, we defined the corresponding change in y as $\Delta y = f(x + \Delta x) - f(x)$ and

$$\frac{dy}{dx} = f'(x) \rightarrow dy = f'(x)dx = f'(x)\Delta x \quad \text{Note that } dy \neq \Delta y \text{ in general.}$$

However, we can think of dy and Δy as approximations of each other, that is $dy \approx \Delta y$.

The diagram above illustrates this.

Note: It is very important to notice that for x -values close to x it is true that $dy \approx \Delta y$. That is to say that, dy and Δy get closer to each other as Δx gets smaller. We show in the next example that the distance between dy and Δy approaches 0 as Δx approaches. We call the difference $\Delta y - dy$ the error.

Exploration: We now want to compare error when using a differential to the amount of change in x .

We can do this by looking at the ratio of these two values and seeing what happens as $\Delta x \rightarrow 0$?

We know that $\Delta y = f(x + \Delta x) - f(x)$ and that $dy = f'(x)\Delta x$. Therefore, we compute:

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{\Delta y - dy}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x) - f'(x)\Delta x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} - \lim_{\Delta x \rightarrow 0} \frac{f'(x)\Delta x}{\Delta x} \\ &= f'(x) - f'(x) \\ &= 0 \end{aligned}$$

Therefore, we see that the change in dy & Δy moves towards 0 as Δx approaches 0.

This shows us that the error is much, much smaller than the change in x when Δx is near 0.

This new notion of differentiability as it relates to behaviors within differentials is what we wish to extend upon.

This is what leads us to define differentiability for functions of two variables as we did above.

Theorem: If a function $f(x, y)$ is differentiable at a point (x_0, y_0) , then it is continuous at that point.

Proof: See page 943 of the textbook.

Note: This is a great result which shows us that we are able to extend the notion of differentiability in the way we desired!

It can be very difficult to use the above definition to determine whether a two or three variable function is differentiable at a point. This leads us to the following more useful theorem.

Theorem: If all first-order partial derivatives of f exist and are continuous at a point, then f is differentiable at that point.

Proof: Beyond the scope of the course.

Note: This theorem applies to functions with any number of variables!

Example 1: Is the function $f(x, y) = x + \sqrt{x + y}$ differentiable at $(1, 1)$.

We see that $f_x(x, y) = 1 + \frac{1}{2\sqrt{x+y}}$ & $f_y(x, y) = \frac{1}{2\sqrt{x+y}}$ and since both of these partial derivatives are continuous at $(1, 1)$ we conclude that $f(x, y)$ is differentiable at $(1, 1)$.